CUSTOMER SEGMENTATION & SHOPPING TRENDS ANALYSIS

A Detailed Analytical Report

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Dataset: 3,900-record consumer shopping trends dataset
Tools: MySQL · Power BI

Executive Summary

This report analyzes 3,900 customer transactions to understand where discounting is most effective, how spend varies across demographics, payment methods, and seasons, and which states generate the highest average order value. The findings are intended to support targeted discount policy, seasonal promotional planning, and regional marketing prioritization.

- Overall average order value across all customers is $59.76, with an average review rating of 3.75.
- Discount rates already scale inversely with average spend across categories, from 42.08% on Clothing up to 44.44% on Outerwear.
- Fall is the strongest season for spend ($61.56 average), while Summer is the weakest ($58.41).
- Alaska, Pennsylvania, and Arizona are the top three states by average purchase value, each exceeding $66.

1. Discount Analysis by Product Category

The table below summarizes average purchase amount, average review rating, transaction volume, and current discount percentage for each product category.

Category	Avg. Purchase (USD)	Avg. Review Rating	Transactions	Discount %
Clothing	$60.03	3.72	1,737	42.08%
Footwear	$60.26	3.79	599	43.24%
Accessories	$59.84	3.77	1,240	43.79%
Outerwear	$57.17	3.75	324	44.44%
Total	$59.76	3.75	3,900	43.00%

Clothing is by far the highest-volume category (1,737 transactions, 44.5% of all orders) yet already carries the lowest discount rate. Outerwear, the lowest-volume category, carries the deepest discount but also the lowest average purchase amount and a below-average order size — suggesting the discount is not yet translating into materially higher basket value.

2. Spending Patterns: Demographics, Payment Method, and Season

Average purchase amount was examined across age band, payment method, and season to identify where spend is concentrated.

2.1 Age Band and Payment Method

Average order values are broadly consistent across all six age bands (18–25 through 66+), generally ranging from the high-$50s to low-$60s. No single age group stands out as dramatically higher-spending, indicating that promotional messaging can largely be age-agnostic. Across payment methods, Bank Transfer, Credit Card, Debit Card, and PayPal cluster near the top of the range within each age band, while Cash trends slightly lower — consistent with cash purchases often being smaller, more discretionary transactions.

Note: this chart does not display printed data labels on the bars, so the figures below are approximate values read directly off the chart (min–max across the six payment methods within each age band) rather than exact underlying figures.

Age Band	Approx. Low (USD)	Approx. High (USD)	Highest-Spend Method(s)
18–25	~$55	~$63	Credit Card / Debit Card / PayPal
26–35	~$60	~$66	Venmo / Debit Card
36–45	~$55	~$63	Debit Card / PayPal
46–55	~$58	~$63	Credit Card / PayPal
56–65	~$58	~$65	Debit Card / PayPal
66+	~$60	~$65	Bank Transfer / Venmo

Across every age band, Cash consistently sits at or near the low end of the range, while Credit Card, Debit Card, PayPal, and Venmo trend toward the high end — reinforcing that encouraging card- and app-based payment could modestly lift average basket size.

2.2 Seasonal Trends

Season	Avg. Purchase (USD)
Fall	$61.56
Winter	$60.36
Spring	$58.74
Summer	$58.41

Spend peaks in Fall and Winter — likely driven by back-to-school and holiday shopping cycles — and dips in Spring and Summer. The gap between the highest (Fall) and lowest (Summer) season is $3.15, or roughly 5% of average order value.

3. Geographic Analysis: Top-Performing States

Average purchase amount by state highlights where customers spend the most per transaction. The top eleven states are shown below.

Rank	State	Avg. Purchase (USD)
1	Alaska	$67.60
2	Pennsylvania	$66.57
3	Arizona	$66.55
4	West Virginia	$63.88
5	Nevada	$63.38
6	Washington	$63.33
7	North Dakota	$62.89
8	Virginia	$62.88
9	Utah	$62.58
10	Michigan	$62.10
11	Tennessee	$61.97

Alaska leads all states at $67.60 average purchase value — nearly $8 above the company-wide average of $59.76. The next tier (Pennsylvania, Arizona) clusters tightly around $66.5, followed by a broader group of states in the low-to-mid $60s.

Key Insights & Recommendations

- Rebalance discounting toward volume: Clothing drives the most transactions but receives the smallest discount — testing a modest increase here may have greater revenue impact than further discounting the already heavily-discounted Outerwear category.
- Time major promotions around Spring and Summer to counteract the seasonal dip, rather than concentrating discount budget in Fall/Winter when spend is already strong.
- Prioritize marketing spend in high-value states (Alaska, Pennsylvania, Arizona) for premium or higher-margin product lines, while using broader campaigns elsewhere to lift average order value.
- Encourage card-based payment methods where feasible, as cash transactions trend toward smaller basket sizes.
- Treat age band as a secondary targeting variable — segment primarily by season, geography, and category rather than age, given the narrow spend variation across age groups.

Methodology

Data was queried and prepared in MySQL from a 3,900-record consumer shopping transactions dataset, then modeled and visualized in Power BI. Metrics shown are aggregate averages and counts computed directly from the underlying transaction table; no statistical significance testing was applied at this stage.

Conclusion

Across category, seasonal, and geographic dimensions, this analysis surfaces clear, actionable patterns: discount allocation can be better aligned with transaction volume, promotional timing should target the Spring/Summer slowdown, and a small set of high-value states merit focused marketing investment. These findings set up the next phase of the project — K-Means clustering to formally segment customers and translate these patterns into targeted marketing personas.